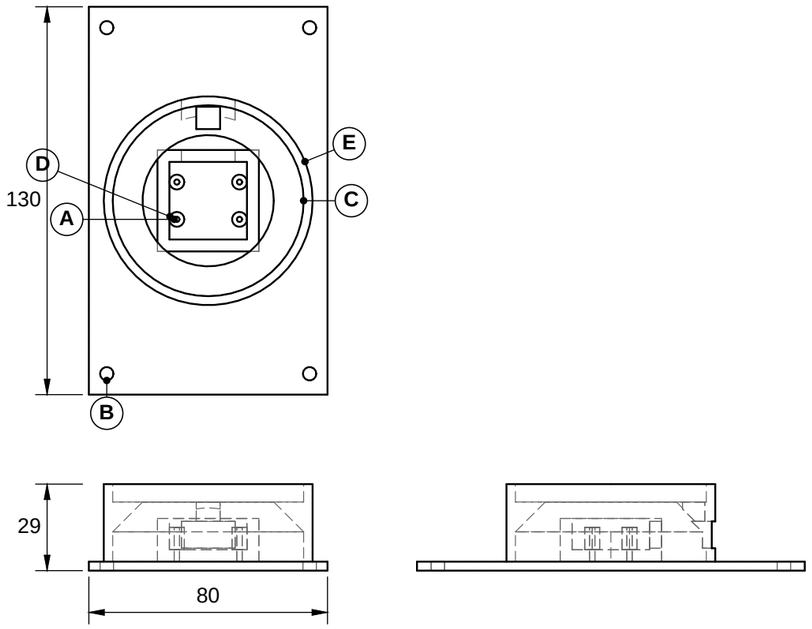
## station_mount

Printed part (P), qty 1. File cad/out/printed/station_mount.step / .stl. Print: PETG black, 3 walls, 25% infill; ribbon slot bridges

SCALE 1:2

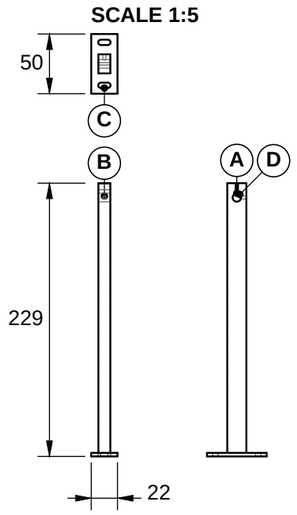



| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | 4x Ø1.8 x 11.5 deep, axis Z, on 21 x 12.5 (X x Y): camera board, M2 self-tapping |
| B | 4x Ø4.5 THRU, axis Z, on 68 x 116 (X x Y): #8 wood screws into the bed |
| C | Ø64 x 26 deep, axis Z: station LED ring seat bore |
| D | 4x Ø5 x 7.5 long (outside), axis Z: M2 boss |
| E | Ø70 x 26 long (outside), axis Z: ring seat tube |
| F | Ring shelf (ring back face, LEDs up) 20 above the base-plate top, which is level with the pad tops |
| G | Camera bosses: tops 11.5 above the base-plate top (board back); lens front STATION_LENS_Z = 23 |

## gate_post_L

Printed part (P), qty 1. File cad/out/printed/gate_post_L.step / .stl. Print: PETG, upright on its foot (as exported), 10 mm brim, 4 walls; needs 230 mm Z height, else print lying on its side with supports

SCALE 1:5

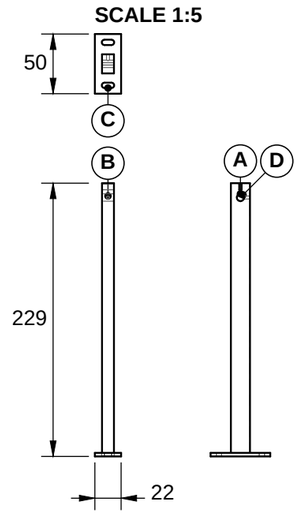



| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | Ø7.2 THRU, axis X: laser barrel bore (7 mm Class 2 laser) |
| B | Ø2.5 THRU, axis Y: M3 set screw onto the barrel (tap or self-form) |
| C | 2x slot 4.5 wide x 10.5 long, axis Z, centres 36 in Y apart: yaw slot, wood screws |
| D | Barrel bore axis 217 above the foot underside: on bare plywood the beam is GATE_Z = 214 above the pad top (BUILD_GUIDE 5.9: no shim) |

## gate_post_R

Printed part (P), qty 1. File cad/out/printed/gate_post_R.step / .stl. Print: PETG, upright on its foot (as exported), 10 mm brim, 4 walls; needs 230 mm Z height, else print lying on its side with supports

SCALE 1:5




| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | Ø6.2 THRU, axis X: receiver barrel bore (6 mm module) |
| B | Ø2.5 THRU, axis Y: M3 set screw onto the barrel (tap or self-form) |
| C | 2x slot 4.5 wide x 10.5 long, axis Z, centres 36 in Y apart: yaw slot, wood screws |
| D | Barrel bore axis 217 above the foot underside: on bare plywood the beam is GATE_Z = 214 above the pad top (BUILD_GUIDE 5.9: no shim) |

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

| | |
|---|---|
| **SlabRail - graded-slab batch scanner** | **SHEET D-07 of D-16** |
| **Printed parts: bed (station mount, gate posts)** | SCALE 1:2, 1:5 (as noted) |
| Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication | UNITS: mm 3RD ANGLE |
| SOURCE: cad/model.py @ 09ca639 (drawn by cad/drawings.py) | DATE 2026-10-27 **DO NOT SCALE** |